```yaml
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name: cmo-urban-last-mile-realism
description: >
  Physics-grounded equation library for urban scenario realism in
  Command: Modern Operations. Use as an analytical overlay, not a
  substitute for internal sensor, weapon, terrain, and damage models.
version: "1.0"
language: en
intended_use:
  - Fictional scenario design and aggregate engagement adjudication.
  - Urban visibility, exposure, sensing, uncertainty, and timing analysis.
  - External spreadsheets, Monte Carlo analysis, and scenario-side logic.
non_goals:
  - Real-world targeting or operational firing solutions.
  - Replacement of authoritative weapons-effects assessment.
  - Assertions that these equations are implemented internally by CMO.
safety_note: >
  Wargaming only. Do not apply to selecting real-world targets or
  operational strike planning.
evidence_note: >
  CMO references support game context, not validation of every formula.
  Physical identities, idealized models, and tunable abstractions are
  distinguished below. Numerical scenario parameters are illustrative.
cmo_context:
  - CMO provides terrain, line-of-sight, sensor, and weapon models.
  - Urban overlays must not double-count existing engine penalties.
  - Coordinate accuracy, identification, hit, and mission effect differ.
units:
  distance: m
  time: s
  angle: rad
  velocity: m/s
  acceleration: m/s^2
  mass: kg
  pressure: Pa
  energy: J
  frequency: Hz
conversions:
  knots_to_mps: "V_mps = 0.514444 * V_kt"
  feet_to_meters: "h_m = 0.3048 * h_ft"
  nautical_miles_to_meters: "R_m = 1852 * R_nm"
  degrees_to_radians: "theta_rad = pi * theta_deg / 180"
constants:
  gravity_mps2: 9.80665
  earth_radius_m: 6371000
  nominal_sea_level_density_kg_m3: 1.225
variables:
  R: Slant range.
  Rg: Ground range.
  Hs: Sensor height above a common local datum.
  Ht: Target height above the same datum.
  Hb: Obstacle-top height above the same datum.
  D: Horizontal observer-to-obstacle distance.
  Dt: Horizontal observer-to-target distance.
  sigma: Per-axis standard deviation unless otherwise stated.
  CEP: Radius enclosing 50 percent of impacts.
  TLE: Target-location error.
  tau: Observation-to-action latency.
  t_f: Weapon flight time.
  r_eff: Modeled effective target radius.
  Pd: Detection probability.
  Pi: Conditional identification probability.
  Ptrack: Conditional track-continuity probability.
  Ph: Conditional geometric hit probability.
  Pk_given_hit: Conditional mission-effect probability after a hit.
  lambda: Wavelength, except lambda_break or lambda_hit.
  rho: Air density.
  W_explosive: TNT-equivalent explosive mass.
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```yaml
assumption_profiles:
  note: >
    All ranges are illustrative scenario inputs, not empirical defaults.
    Building geometry alone cannot determine exposure or sensor performance.
  sparse_suburban:
    building_height_m: "4-12"
    street_width_m: "12-30"
    exposure_probability: "0.35-0.75"
    clutter_factor: "0.10-0.35"
  dense_urban:
    building_height_m: "12-45"
    street_width_m: "8-20"
    exposure_probability: "0.15-0.50"
    clutter_factor: "0.35-0.70"
  high_rise_core:
    building_height_m: "45-200+"
    street_width_m: "8-30"
    exposure_probability: "0.05-0.30"
    clutter_factor: "0.55-0.90"
  industrial_port:
    building_height_m: "6-35"
    street_width_m: "15-60"
    exposure_probability: "0.30-0.70"
    clutter_factor: "0.25-0.65"
equations:
  - id: geo-001
    name: "Slant range"
    category: geometry
    equation: "R = sqrt(Rg^2 + (Hs-Ht)^2)"
    basis: "Flat-local geometry"
    application_and_limits: >
      Compare slant range with sensor range; visibility is
      separate.
  - id: geo-002
    name: "Elevation angle"
    category: geometry
    equation: "theta_e = atan2(Ht-Hs, Rg)"
    basis: "Flat-local geometry"
    application_and_limits: >
      Check viewing angles and obstacle masking.
  - id: geo-003
    name: "Obstacle clearance"
    category: line_of_sight
    equation: "H_LOS(D) = Hs + (Ht-Hs)*(D/Dt); clear if Hb < H_LOS(D)"
    basis: "Idealized straight ray"
    application_and_limits: >
      Use a common elevation datum; test all relevant obstacles.
  - id: geo-004
    name: "Roofline clearance margin"
    category: line_of_sight
    equation: "M_clear = H_LOS(D) - Hb"
    basis: "Geometric identity"
    application_and_limits: >
      Near-zero clearance is sensitive to map and height errors.
  - id: geo-005
    name: "Street-canyon angular sky opening"
    category: geometry
    equation: "F_sky_2D = (2/pi)*atan(W_street/(2*H))"
    basis: "Idealized 2D angular fraction"
    application_and_limits: >
      Not a hemispherical radiative sky-view factor; only a
      geometry proxy.
  - id: geo-006
    name: "Street-canyon aspect ratio"
    category: geometry
    equation: "AR = H/W_street"
    basis: "Geometric ratio"
    application_and_limits: >
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    Increasing AR suggests confinement; thresholds need
    calibration.
- id: geo-007
  name: "Projected planar area"
  category: geometry
  equation: "A_proj = A_face*max(0, cos(phi))"
  basis: "Planar projection"
  application_and_limits: >
    Complex targets require multiple faces or a 3D model.
- id: geo-008
  name: "Visible-area fraction"
  category: visibility
  equation: "f_open = A_visible/A_total"
  basis: "Geometric ratio"
  application_and_limits: >
    Distinguish partial exposure from full target observation.
- id: kin-001
  name: "Flight time"
  category: kinematics
  equation: "t_f = R/v_avg"
  basis: "Constant mean-speed approximation"
  application_and_limits: >
    Prefer CMO flight profiles; maneuvers alter path length.
- id: kin-002
  name: "Target displacement"
  category: kinematics
  equation: "d_move = v_t*(tau+t_f)"
  basis: "Constant velocity"
  application_and_limits: >
    Unknown direction converts displacement into uncertainty,
    not a known offset.
- id: kin-003
  name: "Accelerating displacement"
  category: kinematics
  equation: "d = v_0*t + 0.5*a*t^2"
  basis: "Constant acceleration"
  application_and_limits: >
    Use only over a credible motion interval.
- id: kin-004
  name: "Lateral motion during flight"
  category: kinematics
  equation: "L = v_perp*t_f"
  basis: "Constant lateral velocity"
  application_and_limits: >
    A conceptual moving-target feasibility check, not a firing
    solution.
- id: kin-005
  name: "Relative velocity"
  category: kinematics
  equation: "V_rel = V_weapon - V_target"
  basis: "Vector identity"
  application_and_limits: >
    Closing speed is the appropriate line-of-sight projection.
- id: kin-006
  name: "Dynamic pressure"
  category: aerodynamics
  equation: "q = 0.5*rho*V_air^2"
  basis: "Aerodynamic identity"
  application_and_limits: >
    Use air-relative speed, not weapon-to-target relative
    velocity.
- id: kin-007
  name: "Drag force"
  category: aerodynamics
  equation: "F_D = 0.5*rho*V_air^2*Cd*S"
  basis: "Aerodynamic model"
  application_and_limits: >
    Cd depends on Mach, attitude, and Reynolds number.
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- id: kin-008
  name: "Lift force"
  category: aerodynamics
  equation: "F_L = 0.5*rho*V_air^2*CL*S"
  basis: "Aerodynamic model"
  application_and_limits: >
    Not a substitute for actual weapon control and flight
    performance.
- id: kin-009
  name: "Banked turn radius"
  category: maneuver
  equation: "r_turn = V^2/(g*tan(phi_bank))"
  basis: "Coordinated level turn"
  application_and_limits: >
    Steady flight approximation; not a universal missile
    maneuver law.
- id: kin-010
  name: "Lateral acceleration"
  category: maneuver
  equation: "a_lat = V^2/r_turn"
  basis: "Circular-motion identity"
  application_and_limits: >
    Do not invent airframe acceleration limits.
- id: env-001
  name: "Air-relative velocity"
  category: atmosphere
  equation: "V_air = V_ground - Vwind"
  basis: "Vector identity"
  application_and_limits: >
    Separate wind effects from ground-track motion.
- id: env-002
  name: "Crosswind drift proxy"
  category: atmosphere
  equation: "d_cross ~= V_crosswind*t_f"
  basis: "Passive-advection approximation"
  application_and_limits: >
    Does not model full ballistic drift; guided systems may
    correct it.
- id: env-003
  name: "Density altitude"
  category: atmosphere
  equation: "DA_ft ~= PA_ft + 120*(OAT_C-ISA_temp_C)"
  basis: "Aviation rule of thumb"
  application_and_limits: >
    Approximate; omits explicit humidity and detailed
    atmosphere.
- id: env-004
  name: "Atmospheric transmission"
  category: electro_optical
  equation: "T_atm = exp(-beta*R)"
  basis: "Beer-Lambert attenuation"
  application_and_limits: >
    For uniform extinction; wavelength and smoke matter.
- id: env-005
  name: "Irradiance falloff"
  category: electro_optical
  equation: "E = I/R^2"
  basis: "Point-source inverse-square model"
  application_and_limits: >
    I is radiant intensity; lasers and extended sources require
    other geometry.
- id: env-006
  name: "Diffraction limit"
  category: electro_optical
  equation: "theta_min = 1.22*lambda/Dap"
  basis: "Circular-aperture Rayleigh criterion"
  application_and_limits: >
    Atmosphere, jitter, and clutter often dominate practical
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    resolution.
- id: env-007
  name: "Ground sample distance"
  category: electro_optical
  equation: "GSD ~= R*p/f"
  basis: "Small-angle image-plane approximation"
  application_and_limits: >
    Oblique ground projection is direction dependent; p is pixel
    pitch.
- id: env-008
  name: "Angular target size"
  category: electro_optical
  equation: "alpha ~= L/R"
  basis: "Small-angle approximation"
  application_and_limits: >
    Compare with usable resolution, not just nominal sensor
    range.
- id: env-009
  name: "Pixels across target"
  category: electro_optical
  equation: "n_pixels ~= L/GSD"
  basis: "Sampling approximation"
  application_and_limits: >
    Pixel count alone does not establish identification.
- id: env-010
  name: "Contrast-to-noise"
  category: electro_optical
  equation: "CNR = abs(S_target-S_background)/sigma_noise"
  basis: "Signal separability metric"
  application_and_limits: >
    Urban thermal clutter and reflection can reduce
    separability.
- id: env-011
  name: "Logistic detection model"
  category: sensors
  equation: "Pd = 1/(1+exp(-k*(CNR-CNR_50)))"
  basis: "Tunable abstraction"
  application_and_limits: >
    k and CNR_50 require calibration; not universal sensor
    physics.
- id: env-012
  name: "Detection hierarchy"
  category: sensors
  equation: "Pd >= Pr >= Pi"
  basis: "Nested-event consistency rule"
  application_and_limits: >
    Use comparable unconditional probabilities for this
    inequality.
- id: env-013
  name: "Monostatic radar equation"
  category: radar
  equation: "P_rx = P_tx*Gt*Gr*lambda^2*RCS/((4*pi)^3*R^4*L_loss)"
  basis: "Idealized radar equation"
  application_and_limits: >
    Do not infer urban detection range without propagation and
    processing data.
- id: env-014
  name: "Radar range scaling"
  category: radar
  equation: "R_2/R_1 = (RCS_2/RCS_1)^(1/4)"
  basis: "Fixed-threshold idealized scaling"
  application_and_limits: >
    Other radar and propagation conditions must remain equal.
- id: env-015
  name: "Doppler shift"
  category: radar
  equation: "f_d = 2*v_r/lambda"
  basis: "Monostatic radial-velocity model"
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    application_and_limits: >
      Stationary or tangential targets can be hard to separate
      from clutter.
  - id: env-016
    name: "Sound level attenuation"
    category: acoustics
    equation: "SPL_2 = SPL_1 - 20*log10(R_2/R_1)"
    basis: "Free-field point-source model"
    application_and_limits: >
      Urban reflections, screening, and noise violate assumptions.
  - id: env-017
    name: "First Fresnel-zone radius"
    category: radio_propagation
    equation: "r_F1 = sqrt(lambda*d1*d2/(d1+d2))"
    basis: "Propagation geometry"
    application_and_limits: >
      Geometric line of sight alone does not ensure a clear RF
      path.
  - id: env-018
    name: "Free-space path loss"
    category: radio_propagation
    equation: "FSPL_dB = 20*log10(R_km)+20*log10(f_MHz)+32.44"
    basis: "Free-space model"
    application_and_limits: >
      An optimistic baseline; add urban propagation losses
      separately.
  - id: nav-001
    name: "Combined position uncertainty"
    category: uncertainty
    equation: "sigma_total = sqrt(sum(sigma_i^2))"
    basis: "Independent zero-mean error model"
    application_and_limits: >
      Use consistent per-axis errors; biases and correlations need
      separate treatment.
  - id: nav-002
    name: "CEP to sigma"
    category: uncertainty
    equation: "sigma = CEP/1.177"
    basis: "Isotropic bivariate Gaussian"
    application_and_limits: >
      Not valid for arbitrary anisotropic or biased errors.
  - id: nav-003
    name: "Sigma to CEP"
    category: uncertainty
    equation: "CEP = 1.177*sigma"
    basis: "Isotropic bivariate Gaussian"
    application_and_limits: >
      Sigma denotes the standard deviation of each Cartesian axis.
  - id: nav-004
    name: "Process-noise uncertainty growth"
    category: uncertainty
    equation: "sigma_AoI = sqrt(sigma_0^2 + q_a*t^3/3)"
    basis: "White acceleration-noise model"
    application_and_limits: >
      q_a has units m^2/s^3; initial velocity uncertainty is
      omitted.
  - id: nav-005
    name: "Stale-track uncertainty proxy"
    category: uncertainty
    equation: "sigma_stale ~= sqrt(sigma_0^2+(v_uncertain*tau)^2)"
    basis: "Tunable uncertainty approximation"
    application_and_limits: >
      v_uncertain is an uncertainty-growth proxy, not known target
      speed.
  - id: nav-006
    name: "Angular-to-cross-range error"
    category: uncertainty
    equation: "sigma_cross ~= R*sigma_theta"
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    basis: "Small-angle error propagation"
    application_and_limits: >
      Angles must be in radians.
  - id: nav-007
    name: "Ranging timing error"
    category: uncertainty
    equation: "sigma_range ~= c*sigma_t/2"
    basis: "Round-trip time-of-flight model"
    application_and_limits: >
      c is propagation speed; multipath can introduce bias.
  - id: nav-008
    name: "Uncertainty ellipse area"
    category: uncertainty
    equation: "A_ellipse = pi*a*b"
    basis: "Ellipse identity"
    application_and_limits: >
      State confidence level of semi-axes a and b.
  - id: nav-009
    name: "Circular hit probability"
    category: hit_probability
    equation: "Ph = 1-exp(-r_eff^2/(2*sigma_total^2))"
    basis: "Centered isotropic Gaussian"
    application_and_limits: >
      No aimpoint bias; r_eff is a modeled footprint, not a
      validated effects radius.
  - id: nav-010
    name: "Rectangular hit probability"
    category: hit_probability
    equation: "Ph = erf(L/(2*sqrt(2)*sigma_x))*erf(W/(2*sqrt(2)*sigma_y))"
    basis: "Independent Gaussian axes"
    application_and_limits: >
      Centered aimpoint and aligned rectangle; no axis
      correlation.
  - id: nav-011
    name: "Exposure-window feasibility"
    category: timing
    equation: "feasible = 1 if Texpose >= tau_detect+tau_classify+tau_decide+tau_launch+t_f else 0"
    basis: "Scenario timing gate"
    application_and_limits: >
      Applies where uninterrupted exposure is required;
      reacquisition changes the gate.
  - id: nav-012
    name: "Track continuity"
    category: timing
    equation: "Ptrack = exp(-lambda_break*Ttrack)"
    basis: "Poisson loss-event abstraction"
    application_and_limits: >
      Calibrate loss rate; masks need not occur independently.
  - id: nav-013
    name: "Multiple-look detection"
    category: sensor_fusion
    equation: "P_any = 1-product(1-Pd_i)"
    basis: "Independent-trial probability"
    application_and_limits: >
      Shared masking, weather, or cues invalidate independence.
  - id: nav-014
    name: "Scalar fusion variance"
    category: sensor_fusion
    equation: "1/sigma_fused^2 = sum(1/sigma_i^2)"
    basis: "Independent unbiased scalar estimates"
    application_and_limits: >
      Correlated estimates and vector geometry require covariance
      treatment.
  - id: war-001
    name: "Blast scaled distance"
    category: blast
    equation: "Z = R/W_explosive^(1/3)"
    basis: "Hopkinson-Cranz similarity variable"
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    application_and_limits: >
      Comparison only; urban shielding, reflections, and
      confinement dominate locally.
  - id: war-002
    name: "Equal scaled-distance comparison"
    category: blast
    equation: "R_2 = R_1*(W_2/W_1)^(1/3)"
    basis: "Similarity scaling"
    application_and_limits: >
      Same burst and environmental conditions; not an urban
      damage-radius prediction.
  - id: war-003
    name: "Fragment energy"
    category: fragmentation
    equation: "E_fragment = 0.5*m_fragment*v_fragment^2"
    basis: "Kinetic-energy identity"
    application_and_limits: >
      Energy alone does not predict injury, penetration, or
      functional damage.
  - id: war-004
    name: "Expected fragment intersections"
    category: fragmentation
    equation: "lambda_hit ~= N_frag*A_exposed/(Omega*R^2)"
    basis: "Uniform solid-angle approximation"
    application_and_limits: >
      Not a casualty model; shielding and nonuniform distributions
      are omitted.
  - id: war-005
    name: "At least one fragment intersection"
    category: fragmentation
    equation: "P_any = 1-exp(-lambda_hit)"
    basis: "Poisson approximation"
    application_and_limits: >
      Requires appropriate expected count and event assumptions.
  - id: war-006
    name: "Normal impact velocity"
    category: terminal_geometry
    equation: "V_normal = V*cos(theta_incidence)"
    basis: "Vector projection"
    application_and_limits: >
      Angle measured from surface normal; does not determine
      penetration.
  - id: war-007
    name: "Impact energy"
    category: terminal_geometry
    equation: "E_impact = 0.5*m*V^2"
    basis: "Kinetic-energy identity"
    application_and_limits: >
      Material, projectile construction, fuze, and yaw remain
      important.
  - id: war-008
    name: "Mission-effect chain"
    category: effects
    equation: "P_effect = Pd*Pi_cond*Ptrack_cond*Ph_cond*Pk_given_hit"
    basis: "Conditional probability chain"
    application_and_limits: >
      Use conditional terms; multiplying marginal probabilities
      needs extra assumptions.
  - id: war-009
    name: "Multiple-attempt effect"
    category: effects
    equation: "P_effect_N = 1-product(1-P_effect_i)"
    basis: "Independent-trial probability"
    application_and_limits: >
      Shared coordinates or target misidentification cause
      correlated failures.
  - id: urban-001
    name: "Visibility penalty proxy"
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    category: urban_abstraction
    equation: "P_visible = P_base*(1-f_mask)*(1-f_occ)*(1-f_clutter)"
    basis: "Tunable abstraction"
    application_and_limits: >
      Factors overlap; avoid double counting and calibrate to
      scenario behavior.
  - id: urban-002
    name: "Opportunity probability proxy"
    category: urban_abstraction
    equation: "P_engageable = p_exp*P_visible*Ptrack*feasible"
    basis: "Scenario abstraction"
    application_and_limits: >
      Factors may be dependent; not a validated universal
      prediction.
  - id: urban-003
    name: "Building shadow length"
    category: urban_geometry
    equation: "L_shadow = H/tan(alpha_sun)"
    basis: "Flat-ground solar geometry"
    application_and_limits: >
      Visible-light shadows do not directly specify thermal
      contrast.
  - id: urban-004
    name: "Temperature contrast"
    category: urban_thermal
    equation: "Delta_T = abs(T_target-T_background)"
    basis: "Thermal contrast proxy"
    application_and_limits: >
      Radiance also depends on emissivity, band, atmosphere, and
      geometry.
  - id: urban-005
    name: "Elevation-induced ground error"
    category: urban_geolocation
    equation: "sigma_ground ~= sigma_h/tan(theta_depression)"
    basis: "Local planar error propagation"
    application_and_limits: >
      Shallow depression angles amplify height uncertainty.
  - id: urban-006
    name: "Oblique-image displacement"
    category: urban_geolocation
    equation: "d_parallax ~= H_building*tan(theta_off_nadir)"
    basis: "Local geometric proxy"
    application_and_limits: >
      Requires elevation-aware registration for reliable target
      mapping.
  - id: urban-007
    name: "Urban radio link budget"
    category: urban_rf
    equation: "P_rx_dBm = P_tx_dBm+G_tx+G_rx-FSPL_dB-L_building_dB-L_misc_dB"
    basis: "Link-budget identity"
    application_and_limits: >
      Gains in dBi; losses in dB; material and multipath effects
      are variable.
  - id: urban-008
    name: "Exposed-route dwell time"
    category: urban_timing
    equation: "T_dwell = L_exposed_route/v_target"
    basis: "Constant-speed approximation"
    application_and_limits: >
      Compare with observation and action latency; stops and turns
      alter dwell.
  - id: urban-009
    name: "Street crossing time"
    category: urban_timing
    equation: "T_cross = W_street/v_target"
    basis: "Perpendicular constant-speed crossing"
    application_and_limits: >
      Use actual crossing path for angled routes.
```

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  - id: urban-010
    name: "Location-quality score"
    category: urban_abstraction
    equation: "Q_TLQ = 1/(1+sigma_total/r_eff)"
    basis: "Heuristic normalized score"
    application_and_limits: >
      Not a probability or internal CMO mechanic; useful only as a
      documented score.
workflow:
  - step: 1
    name: Define target geometry and posture
    actions:
      - Record target footprint, height, aspect, route, and motion state.
      - Separate exposed, masked, rooftop, facade, indoor, and covered targets.
  - step: 2
    name: Represent urban geometry
    actions:
      - Record building elevations, street widths, rooflines, and obstacles.
      - Test visibility by axis and retain uncertainty in incomplete mapping.
  - step: 3
    name: Assess sensing and timing
    actions:
      - Separate detection, recognition, and identification.
      - Compare exposure duration with total observation-to-effect delay.
      - Expand stale-track error and assess reacquisition requirements.
  - step: 4
    name: Assess terminal uncertainty
    actions:
      - Combine independent random errors consistently; retain biases.
      - Evaluate modeled footprint probability, not accuracy in isolation.
      - Account for weather, viewing geometry, and guidance limitations.
  - step: 5
    name: Assess mission effect
    actions:
      - Separate a hit from functional defeat.
      - Prefer CMO database behavior for in-game weapon-target effects.
      - Treat free-field blast and fragment estimates as comparisons only.
  - step: 6
    name: Implement and validate
    actions:
      - Use engine terrain, sensors, doctrine, weather, and target posture first.
      - Add documented event, scoring, or external overlays only where needed.
      - Run sensitivity checks and avoid duplicating engine penalties.
cmo_design_mapping:
  exposed:
    treatment: Normal engine behavior unless evidence warrants an overlay.
  partially_masked:
    treatment: Intermittent visibility and classification or tracking delays.
  deep_urban_masked:
    treatment: Reacquisition requirements and greater stale-track uncertainty.
  indoor_or_hardened:
    treatment: Separate exterior impact from penetration and functional effect.
weapon_guidance_comparison:
  coordinate_guided:
    suitable_context: Fixed, accurately located fictional scenario targets.
    limitations: Location bias, stale coordinates, height error, and movement.
  imaging_terminal:
    suitable_context: Distinguishable targets with sufficient observation.
    limitations: Contrast, obscurants, masking, aspect, and confusion.
  laser_guided:
    suitable_context: Persistent designation and favorable viewing geometry.
    limitations: Line of sight, smoke, weather, and designation continuity.
  radar_guided:
    suitable_context: Some visually obscured conditions, system dependent.
    limitations: Clutter, multipath, masking, aspect, and processing limits.
  unguided:
    suitable_context: Broad fictional targets where engine effects support it.
    limitations: Delivery uncertainty, weather, motion, and observation quality.
```

```yaml
recommended_inputs:
  geometry:
    - Sensor and target position, elevation, dimensions, and orientation.
    - Representative building heights, street widths, and obstruction geometry.
  sensing:
    - Modality, range, viewing angle, resolution, weather, and illumination.
    - Separate detection, recognition, and identification confidence.
  uncertainty:
    - Initial location error, track age, motion uncertainty, and cue latency.
    - Correlation, systematic bias, and reacquisition assumptions.
  weapon:
    - Database guidance type, flight time, accuracy, and relevant limitations.
  effects:
    - Target vulnerability, cover, mission-effect definition, and uncertainty.
example:
  title: Illustrative moving vehicle in a fictional dense urban corridor
  inputs:
    exposed_route_m: 60
    target_speed_mps: 8
    cue_latency_s: 4
    flight_time_s: 12
    initial_sigma_m: 5
    uncertainty_growth_speed_mps: 4
    building_height_m: 25
    street_width_m: 12
  calculations:
    dwell_time_s: 7.5
    total_delay_s: 16
    canyon_aspect_ratio: 2.0833
    stale_sigma_m_approx: 64.195
    uninterrupted_exposure_gate: fails
  interpretation: >
    Exposure ends before modeled arrival. Without continued tracking or
    reacquisition, nominal delivery precision does not resolve the problem.
    This example is a timing and uncertainty illustration, not a prediction
    of a specific CMO weapon or engagement outcome.
validation:
  - Confirm database behavior before applying overlays.
  - Check meters, feet, knots, nautical miles, degrees, and radians.
  - Distinguish ground range, slant range, and obstacle range.
  - Use a common elevation datum for buildings, observer, and target.
  - Separate random errors, systematic biases, and misidentification.
  - State independence assumptions and ellipse confidence levels.
  - Test sensitivity to track age, motion, exposure, and continuity.
  - Use distributions or ranges where inputs are poorly known.
  - Do not infer indoor functional damage from a generic exterior hit.
  - Do not equate the heuristic quality score with a probability.
limitations:
  - No access to or reproduction of CMO internal source code is asserted.
  - Dense urban physics is three-dimensional and often non-Gaussian.
  - Simplified propagation and damage equations omit important local effects.
  - Engine capability and database behavior vary with edition and version.
  - Legal and civilian-protection assessment is not supplied by these models.
references:
  - title: Command Modern Operations Manual
    url: https://ftp.matrixgames.com/pub/CommandModernOperations/CMO%20manual%20EBOOK.pdf
    citation: "[web:14]"
    scope: CMO line-of-sight and game-mechanics context.
  - title: Command Modern Operations Ground Ops discussion
    url: https://forums.matrixgames.com/viewtopic.php?t=340838
    citation: "[web:5]"
    scope: Urban terrain and ground-operations context.
  - title: Combined Arms Operations in Urban Terrain
    url: https://www.bits.de/NRANEU/others/amd-us-archive/fm3-06.11(02).pdf
    citation: "[web:11]"
    scope: General urban terrain constraints, not equation validation.
  - title: No-Strike and Collateral Damage Estimation Methodology
    citation: "[web:2]"
```

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    scope: Context only; this skill is not an authoritative damage methodology.
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```